

Q9: In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Introduction

Computer Vision enables computers to acquire, process, analyze, and understand images. In a facial recognition system, different levels of Computer Vision work together to identify a person accurately. The process begins with image acquisition and ends with intelligent decision-making.

Levels of Computer Vision in Facial Recognition

1. Low-Level Computer Vision (Image Acquisition and Preprocessing)

Low-level computer vision focuses on capturing and improving the image quality.

Functions:

- Capture the face image using a camera.
- Convert the image into digital format.
- Remove noise from the image.
- Adjust brightness and contrast.
- Convert the image to grayscale if required.

Role:

- Produces a clear and high-quality image for further processing.

2. Mid-Level Computer Vision (Feature Extraction)

Mid-level computer vision extracts important information from the image.

Functions:

- Detect the face in the image.
- Locate facial features such as:
 - Eyes

- Nose
 - Mouth
 - Facial boundaries
- Extract unique facial features (feature vectors).

Role:

- Converts the facial image into numerical features that can be compared with stored data.

3. High-Level Computer Vision (Recognition and Decision Making)

High-level computer vision interprets the extracted features and makes intelligent decisions.

Functions:

- Compare extracted facial features with the database.
- Identify or verify the person's identity.
- Decide whether the person is authorized or unauthorized.
- Generate actions such as granting or denying access.

Role:

- Performs recognition and decision making based on the analyzed information.

Working Process of Facial Recognition

1. Image Acquisition

- A camera captures the person's face.

2. Image Enhancement

- Noise is removed and brightness/contrast are adjusted.

3. Face Detection

- The system detects the face in the captured image.

4. Feature Extraction

- Important facial features such as eyes, nose, and mouth are extracted.

5. Face Recognition

- The extracted features are compared with stored facial data.

6. Decision Making

- If a match is found, the person's identity is confirmed.
- Otherwise, access is denied or an alert is generated.

Applications

- Smartphone face unlock
- Attendance management systems
- Airport security
- Banking authentication
- Smart home access control
- Surveillance systems

Conclusion

Facial recognition involves all three levels of Computer Vision. Low-level vision acquires and enhances the image, mid-level vision detects the face and extracts important features, and high-level vision recognizes the person and makes intelligent decisions. Together, these levels enable accurate and reliable facial recognition in real-world applications.